Shafayet Khan Shafee

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Data Scientist specializing in causal inference, machine learning, and experimentation for product decisions, with hands-on experience in fintech, including BNPL systems. Brings end-to-end ownership from framing ambiguous product questions to measurable, actionable outcomes through experimentation and causal and predictive modeling, with additional experience of shipping ML and data pipelines to production. Drawn to problems at the intersection of causal inference, ML, and Bayesian methods, with a strong interest in high-scale, product-led data environments.

Professional Experience

Data Scientist — Dhaka, Bangladesh
Pathao Pay — Jan 2025–Present

- Architected a centralised analytics data layer using dbt and BigQuery on GCP, standardising metric definitions across teams and eliminating ad-hoc query duplication, accelerating BI reporting cycles.
- Designed and owned end-to-end A/B experiments for product and marketing initiatives — including randomization, cohort design, covariate balance assessment, and significance testing — directly informing decisions on user incentive structures and campaign strategies.
- Applied Bayesian experimental design (Bayes Factor Design Analysis) to estimate sample sizes for controlled rollouts, replacing power-based stopping rules with evidence-based criteria and enabling earlier, more confident product decisions.
- Estimated causal impact of a fintech product rollout on merchant transaction behavior using propensity score matching and Difference-in-Differences, enabling fair comparison across treated and control groups.
- Applied Hierarchical Bayesian modeling to evaluate product adoption across cohorts of unequal and small sizes, quantifying uncertainty through posterior distributions to support risk-aware product decisions.

Data Scientist — Dhaka, Bangladesh
Pathao Limited — Jul 2023–Dec 2024

- Built automated end-to-end reporting infrastructure using dbt, BigQuery, Looker Studio, and Google Apps Script, enabling real-time tracking of key business metrics and reducing manual reporting overhead.
- Designed and analyzed randomized controlled experiments to evaluate product initiatives, including randomization checks through covariate balance assessment and Sample Ratio Mismatch (SRM) testing.
- Applied causal ML methods — Double ML and generalized random forests — to estimate heterogeneous treatment effects across user subgroups, enabling targeted intervention design and personalized offer optimization.
- Co-developed a repayment risk scoring framework using Random Survival Forests to model time-to-repayment distributions, guiding targeted outreach to high-risk users and improving CX resource allocation.
- Co-developed a default risk prediction model for new BNPL applicants, enabling risk-stratified cohort rollouts and directly contributing to reduction in non-performing loan (NPL) rates.

Education

M.Sc. in Applied Statistics — Dhaka, Bangladesh
GPA 3.97 / 4.00 — 2022–2023
Institute of Statistical Research & Training
University of Dhaka

B.Sc. in Applied Statistics — Dhaka, Bangladesh
CGPA 3.96 / 4.00 — 2018–2022
Institute of Statistical Research & Training
University of Dhaka

Publications

[1] **S. K. Shafee** and M. S. Rahman. *On the estimation of the median odds ratio for measuring contextual effects in multilevel binary data from complex survey designs*. [Under review, 2026]. arXiv: 2606.15145 [stat.ME].

[2] **S. K. Shafee** et al. *G-computation for causal effect estimation from observational hierarchical data with unmeasured cluster context*. [Under review, 2026]. arXiv: 2606.14131 [stat.ME].

[3] **S. K. Shafee** et al. *Investigating the causal effect of maternal continuum of care on child's minimum acceptable diet: A multilevel approach using partially pooled propensity score weighting*. PLOS ONE, 2025. DOI: 10.1371/journal.pone.0335972.

Awards & Achievements

- **Dean's Award**, *Faculty of Science, University of Dhaka* **2025**
- **Conference Award for Scientists**, *45th Annual Conference of the ISCB, Thessaloniki, Greece* **2024**
- **Datathon Finalist**, Top 10 of 300 teams; built a purchase-behavior prediction model for mobile data packages, *Robi Axiata Limited, Dhaka, Bangladesh.* **2024**
- **National Science and Technology (NST) Fellowship**, M.Sc. thesis research in multilevel modeling, *Ministry of Science and Technology (MoST), Government of Bangladesh* **2023**

Technical Skills

- **Programming Languages:** Python, R, SQL.
- **Machine Learning:** Ensemble Methods, Survival Modeling, Conformal Prediction, Model Calibration, Causal ML (DML, Causal Forest) .
- **Statistics & Experimentation:** Hypothesis Testing, Bayesian Modeling, Causal Inference (IPW, Matching, DiD), A/B Testing, Time Series Analysis.
- **Data & ML Engineering:** dbt, Kedro, MLflow, Bash Scripting.
- **Tools & Platforms:** BigQuery, GCP, Docker, Git, CI/CD (GitHub Actions, GitLab CI).
- **Analytics & Visualization:** Looker Studio, MixPanel, Shiny.

Open Source

- `skmiscpy` — Python package for causal inference diagnostics, including Love plots, mirror histograms, and SMD computation for IPW workflows.
- `randomizer` — Dockerized Shiny app for experimental unit randomization with SRM testing and covariate balance diagnostics.
- `kedrogen` — CLI tool for scaffolding reproducible data science projects from cookiecutter templates.
- `python-uv-gcloud` — Minimal Docker base image with Python, `uv`, and Google Cloud SDK for CI/CD pipelines and cloud workflows.
- `MOR` — R package implementing post-estimation functions to compute the Median Odds Ratio and confidence intervals from fitted multilevel binary logistic models.